

Hindalco Summer Internship Project



**Internship/Academic
Project Report 2026**

IT DEPARTMENT

AI-Based Scrap Management System for Hindalco Industries

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Abstract

The AI-Based Scrap Management System is a web-based application designed to automate the process of scrap tracking, inventory management, recycling monitoring, and reporting within Hindalco Industries. The system aims to reduce manual record-keeping, improve operational efficiency, and support sustainable manufacturing practices.

The application provides real-time monitoring of scrap generation across different departments, tracks recycling activities, generates analytical reports, and uses Machine Learning techniques to predict future scrap generation trends. This helps management make data-driven decisions and optimize resource utilization.

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AI-Based Scrap Management System for Hindalco Industries

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1. Executive Summary

The AI-Based Scrap Management System is an intelligent industrial waste management platform developed for Hindalco Industries to automate scrap tracking, inventory management, recycling operations, reporting, and predictive analytics.

The system replaces manual record-keeping methods with a centralized digital platform that provides real-time visibility into scrap generation across multiple departments. Additionally, Machine Learning algorithms are integrated to forecast future scrap generation patterns, helping management optimize production planning, reduce waste, and improve recycling efficiency.

1.1 Company Overview

About Hindalco Industries

[Hindalco Industries](#) is the metals flagship company of the Aditya Birla Group and one of the world's largest producers of aluminium and copper.

The company operates large-scale manufacturing facilities where significant quantities of industrial scrap are generated during production processes such as:

- Rolling Operations
- Casting Operations
- Extrusion Processes
- Fabrication Activities
- Machining Operations

Efficient management of this scrap is critical for operational efficiency, cost reduction, sustainability, and regulatory compliance.

1.2 Problem Statement

Currently, many manufacturing facilities rely on spreadsheets, paper records, or fragmented systems to manage scrap-related data.

This creates several challenges:

1.3 Operational Challenges

- Manual data entry errors
- Duplicate records
- Lack of centralized monitoring
- Delayed reporting
- Difficult inventory tracking

1.4 Management Challenges

- Poor visibility of department-wise scrap generation
- Difficulty in identifying waste trends
- Limited forecasting capability
- Inefficient recycling planning

1.5 Business Challenges

- Increased operational costs
- Lower recycling efficiency
- Reduced productivity
- Delayed decision-making

2.Project Objectives

The primary objectives of the project are:

➤ Operational Objectives

- Digitize scrap management processes
- Automate inventory tracking
- Streamline recycling operations
- Improve reporting accuracy

➤ Business Objectives

- Reduce scrap handling costs
- Improve resource utilization
- Increase recycling efficiency
- Support sustainable manufacturing

➤ Technical Objectives

- Develop a scalable web application
- Implement role-based access control
- Generate analytical reports
- Integrate AI-powered forecasting

Existing System Analysis

Manual Registers

+

Excel Sheets

+

Department Records

Limitations

- ❖ Human errors
- ❖ Time-consuming reporting
- ❖ Data inconsistency
- ❖ No predictive analysis
- ❖ Poor scalability

Full-Stack Architecture Layout

- **Presentation Tier:** Lightweight frontend engine utilizing standard **HTML5**, **CSS3**, and **JavaScript (ES6+)** with **Fetch API** for high-speed micro-interactions.
- **Application Server (server.js):** Modular backend built on **Node.js** and **Express.js** to handle operational REST endpoints, middleware routes, and request validation.
- **Data Persistence Layer: MongoDB** serving as a NoSQL document database, allowing flexible, schema-less handling of rapid scrap variations.
- **Intelligence Pipeline: Python Engine** integrated with Node.js to consume data clusters and stream algorithmic predictions via JSON payloads.
- **Visual Anchor:** A modern 4-tier architectural flow chart mapping: *Web Client (HTML/JS) → REST Client → Express Backend Server (server.js <-> MongoDB Engine <-> Python ML Core.*

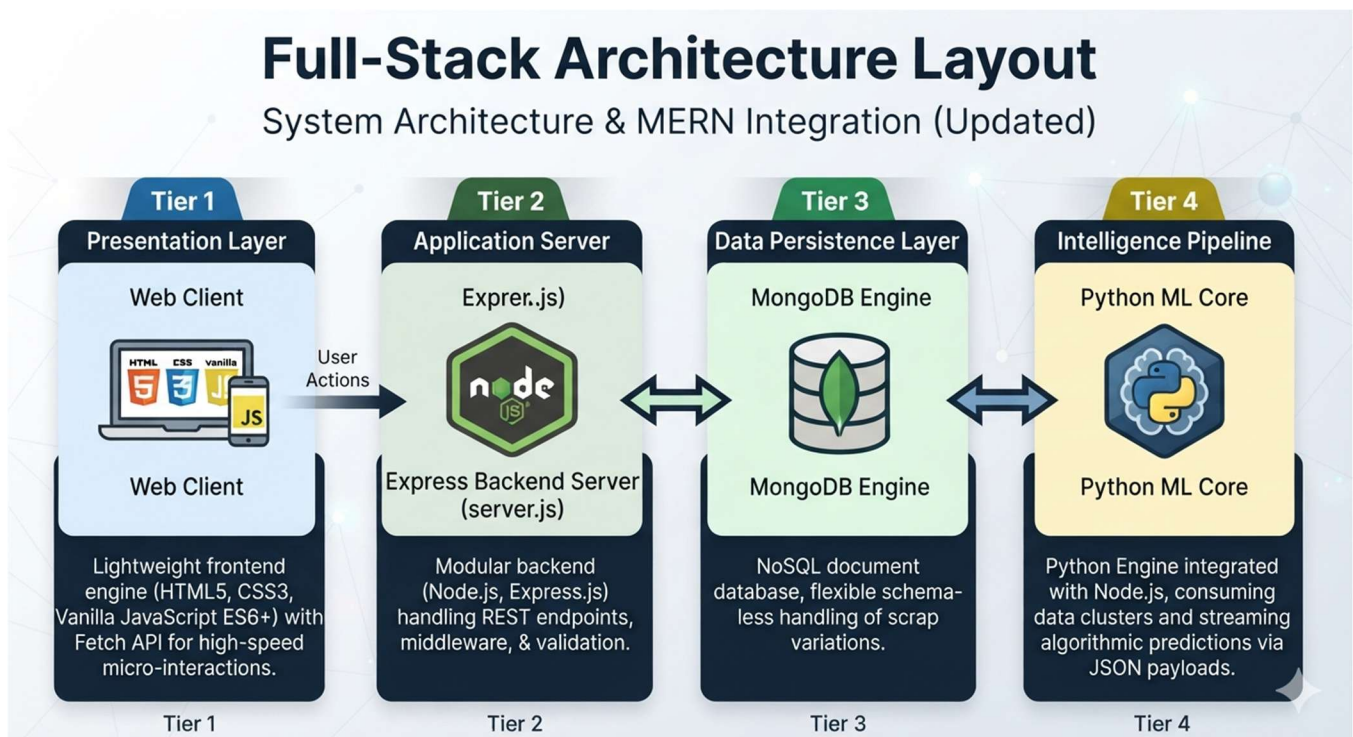
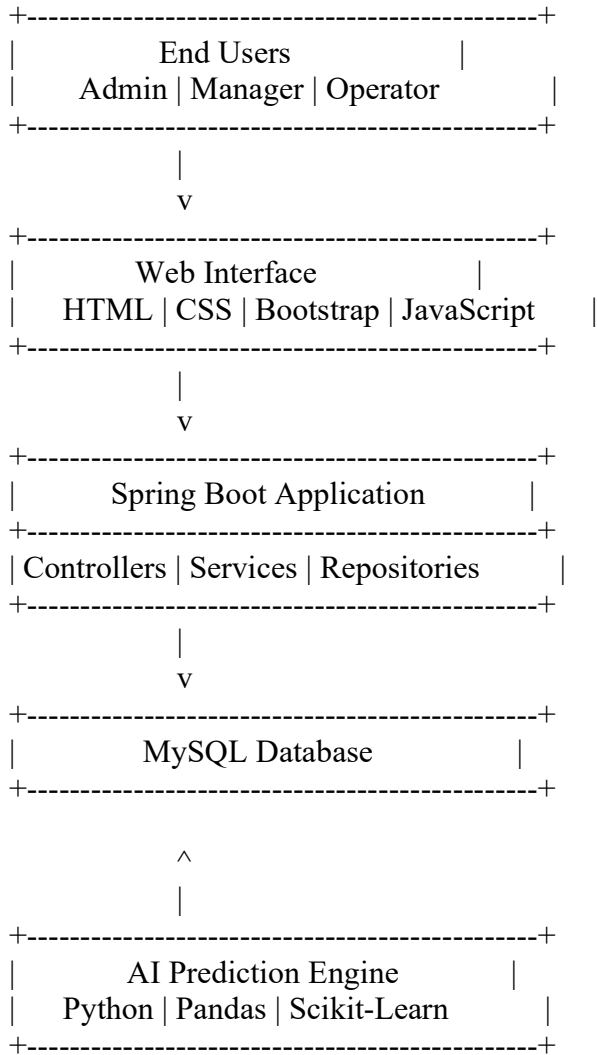


Figure: System Architecture

User → Frontend → Controller → Service → Repository → MongoDB
→ AI Processing → Insights/Recommendations → Frontend → User

System Architecture



System Use Case Diagram

AI SCRAP MANAGEMENT SYSTEM

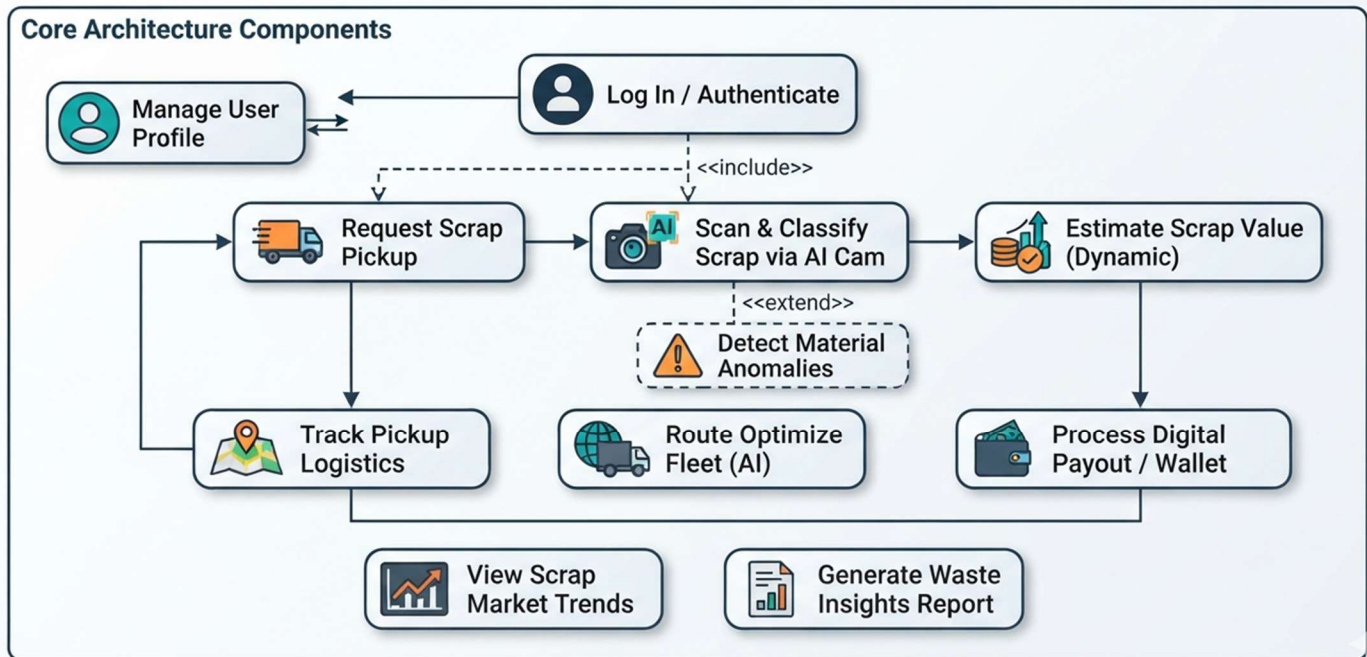


Figure: System Use Case

The System Use Case Diagram represents the major functionalities and interactions within the AI Scrap Management System. It demonstrates how AI technologies, logistics management, scrap classification, valuation, reporting, and user services are integrated into a single intelligent platform for efficient scrap management and operational optimization.

Key Functionalities of the AI Scrap Management System

- Secure user authentication and login management.
- User profile management and account handling.
- Scrap pickup request generation and monitoring.
- Real-time tracking of scrap pickup logistics.
- AI-based scrap scanning and classification.
- Automatic detection of material anomalies.
- Dynamic scrap value estimation based on material type.
- AI-powered route optimization for efficient transportation.
- Digital payout and wallet management.
- Scrap market trend analysis and monitoring.
- Waste insights and analytics report generation.
- Centralized platform for scrap management operations.
- Improved decision-making through AI-driven analytics.
- Enhanced operational efficiency and resource utilization.
- Support for sustainable waste management practices.
-

Database Schema Design Summary Chart

Entity/Table	Primary Key	Foreign Key	Main Attributes	Purpose
Department	department_id	—	department_name, manager_name, location	Stores information about manufacturing departments
Scrap	scrap_id	department_id	scrap_type, weight, scrap_date, remarks	Tracks scrap generated during production
Inventory	inventory_id	scrap_id	available_stock, last_updated	Monitors available scrap inventory
Recycling	recycle_id	scrap_id	recycled_weight, recycle_date, recycling_status	Tracks recycling operations and status
User	user_id	—	full_name, email, password, role	Manages system users and access control
Report	report_id	user_id	report_type, generated_date	Stores generated reports
Prediction	prediction_id	—	prediction_month, predicted_scrap_weight, high_waste_department	Stores AI-based prediction results
Audit_Log	log_id	user_id	action, timestamp	Records user activities and system changes

Database Flow Diagram

Department



Scrap



Recycling



Inventory

User



Report

Audit_Log

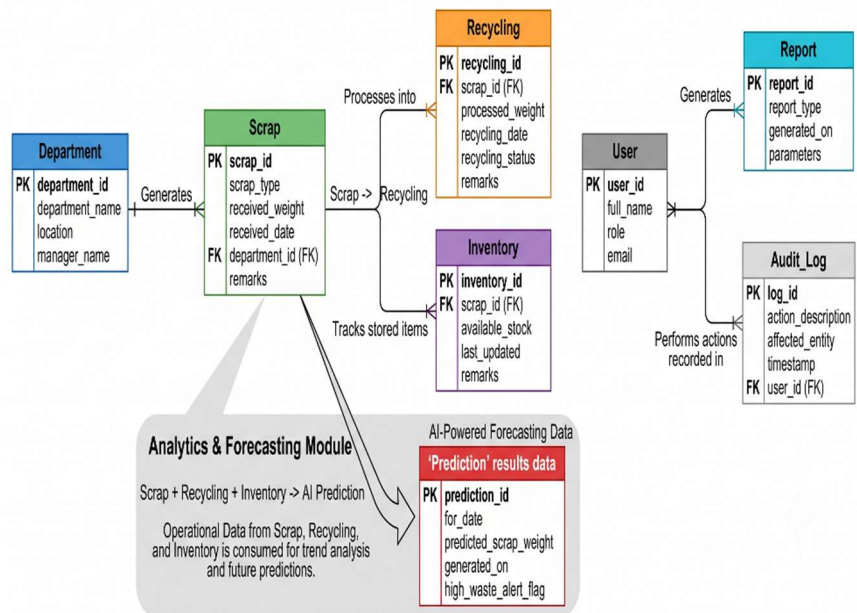
Scrap + Recycling + Inventory



AI Prediction

Enhanced ERD: Industrial Scrap Management and Recycling Optimization with Analytics

Refined Schema for Real-time Monitoring and AI Prediction



METHODOLOGY

3.1 Project Planning and Requirement Analysis

The project begins with a detailed analysis of scrap management practices within aluminium manufacturing operations. The primary objective is to identify inefficiencies associated with scrap tracking, inventory control, recycling processes, and waste analysis. Based on the identified requirements, a centralized digital solution is proposed to automate and optimize scrap management activities.

3.2 System Architecture Design

The system follows a modular and scalable architecture to ensure maintainability, reliability, and future extensibility. The application is designed using a layered architecture consisting of Presentation, Business Logic, Data Access, and Database layers.

The major functional modules include:

- User Authentication and Authorization
- Scrap Entry and Monitoring
- Scrap Categorization
- Inventory Management
- Recycling Management
- Analytics and Reporting
- AI-Based Prediction Module

This architecture enables efficient communication between different system

3.3 Database Design and Management

A relational database is developed using MySQL to securely store and manage operational data. The database schema is designed using normalization principles to eliminate redundancy and maintain data integrity.

The primary entities include:

- Department
- Scrap
- Inventory
- Recycling
- User
- Report
- Prediction

Primary Keys, Foreign Keys, Constraints, and Indexing techniques are implemented to ensure data consistency, security, and optimized query performance.

3.4 Application Development

The web application is developed using Java Spring Boot following industry-standard software engineering practices.

Technology Stack:

- Frontend: HTML5, CSS3, JavaScript, Bootstrap
- Backend: Java Spring Boot
- Database: MySQL
- Visualization: Chart.js
- Machine Learning: Python, Pandas, NumPy, Scikit-Learn

The backend is implemented using Controller, Service, Repository, and Entity layers to ensure a structured and maintainable codebase.

3.5 Scrap Monitoring and Inventory Management

The Scrap Management Module records all scrap generated during manufacturing operations. Relevant information such as scrap type, weight, department, generation date, and remarks is captured and stored within the database.

The Inventory Management Module continuously updates available scrap stock levels and provides real-time visibility of inventory status across departments.

3.6 Recycling Management

The Recycling Management Module monitors scrap recycling activities and maintains records of recycled and pending materials. The module enables performance evaluation through recycling efficiency metrics.

The recycling rate is calculated using the following formula:

$$\text{Recycling Rate (\%)} = (\text{Recycled Scrap} \div \text{Total Scrap Generated}) \times 100$$

The generated metrics assist management in evaluating recycling performance and identifying opportunities for process improvement.

3.7 Artificial Intelligence-Based Prediction

An AI-powered prediction module is incorporated to analyse historical scrap generation data and forecast future trends.

The machine learning workflow consists of:

1. Data Collection
2. Data Preprocessing
3. Feature Engineering
4. Model Training
5. Model Evaluation
6. Prediction Generation

Algorithms such as Linear Regression and Random Forest are utilised to estimate future scrap generation and identify departments contributing significantly to waste production.

The predictive insights support proactive decision-making and operational planning

3.7 Artificial Intelligence-Based Prediction

An AI-powered prediction module is incorporated to analyse historical scrap generation data and forecast future trends.

The machine learning workflow consists of:

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10. Model Training
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Algorithms such as Linear Regression and Random Forest are utilised to estimate future scrap generation and identify departments contributing significantly to waste production.

The predictive insights support proactive decision-making and operational planning.

3.8 Analytics and Reporting

The system provides comprehensive reporting and visualization capabilities to facilitate data-driven decision-making.

Generated reports include:

- Daily Reports
- Weekly Reports
- Monthly Reports
- Annual Reports

Interactive dashboards display:

- Total Scrap Generated
- Recycled Scrap Quantity
- Pending Scrap Quantity
- Recycling Efficiency
- Department-wise Waste Analysis
- Predicted Scrap Trends

Visual representations are provided through bar charts, pie charts, and line charts for enhanced analytical interpretation.

3.9 System Testing and Validation

Comprehensive testing is conducted to ensure system reliability and functional correctness.

The testing process includes:

- Functional Testing
- Database Validation
- User Authentication Testing
- Report Verification
- Prediction Accuracy Assessment
- Performance Evaluation

Any identified defects are corrected prior to deployment to ensure stable and efficient operation.

3.10 Expected Outcomes

The proposed system delivers a centralized and intelligent platform for scrap management and recycling optimization. The implementation of the system is expected to:

- Improve scrap traceability and visibility
- Enhance recycling efficiency
- Reduce material losses and operational costs
- Strengthen inventory control mechanisms
- Support data-driven managerial decisions
- Promote sustainable manufacturing practices
- Increase overall operational efficiency

System Walkthrough & Screenshots

1. Login Page

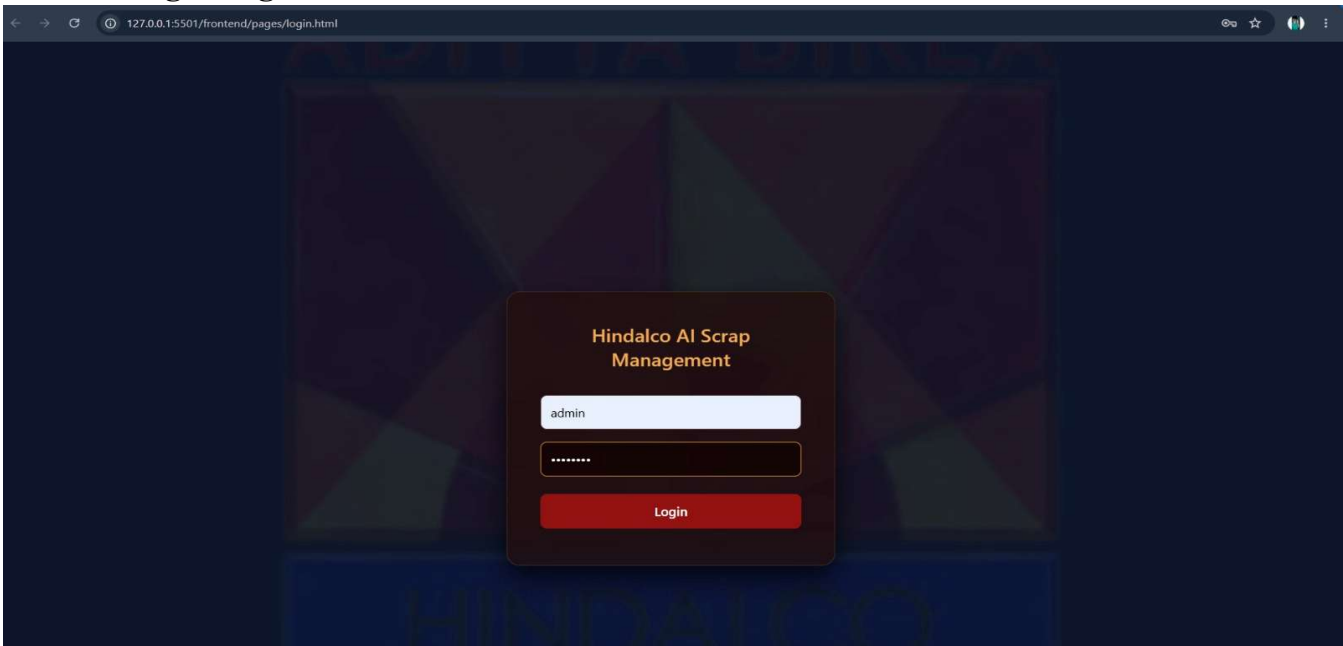


Figure: Login Page

This screen provides secure user authentication for accessing the Scrap Management System. Users enter their credentials to log in and access authorized modules based on their assigned roles.

2. Dashboard

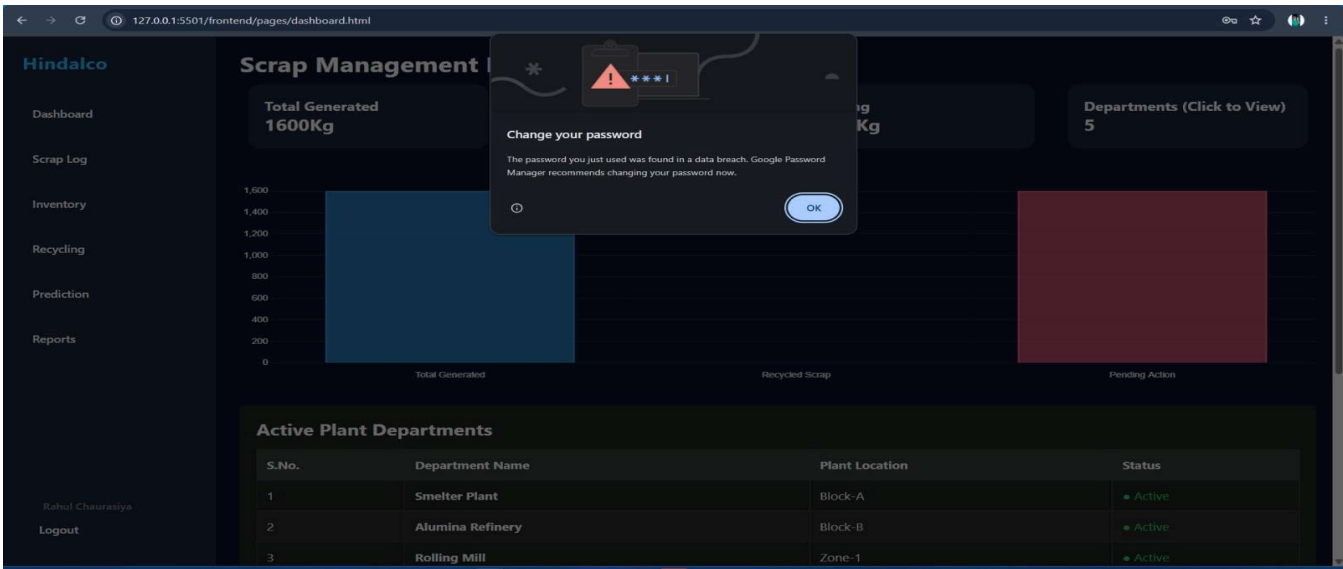


Figure: Dashboard

The dashboard provides a centralized overview of key system metrics, including total scrap generated, recycled scrap, inventory status, and operational insights. It enables quick monitoring and decision-making.

3. Scrap Entry Module

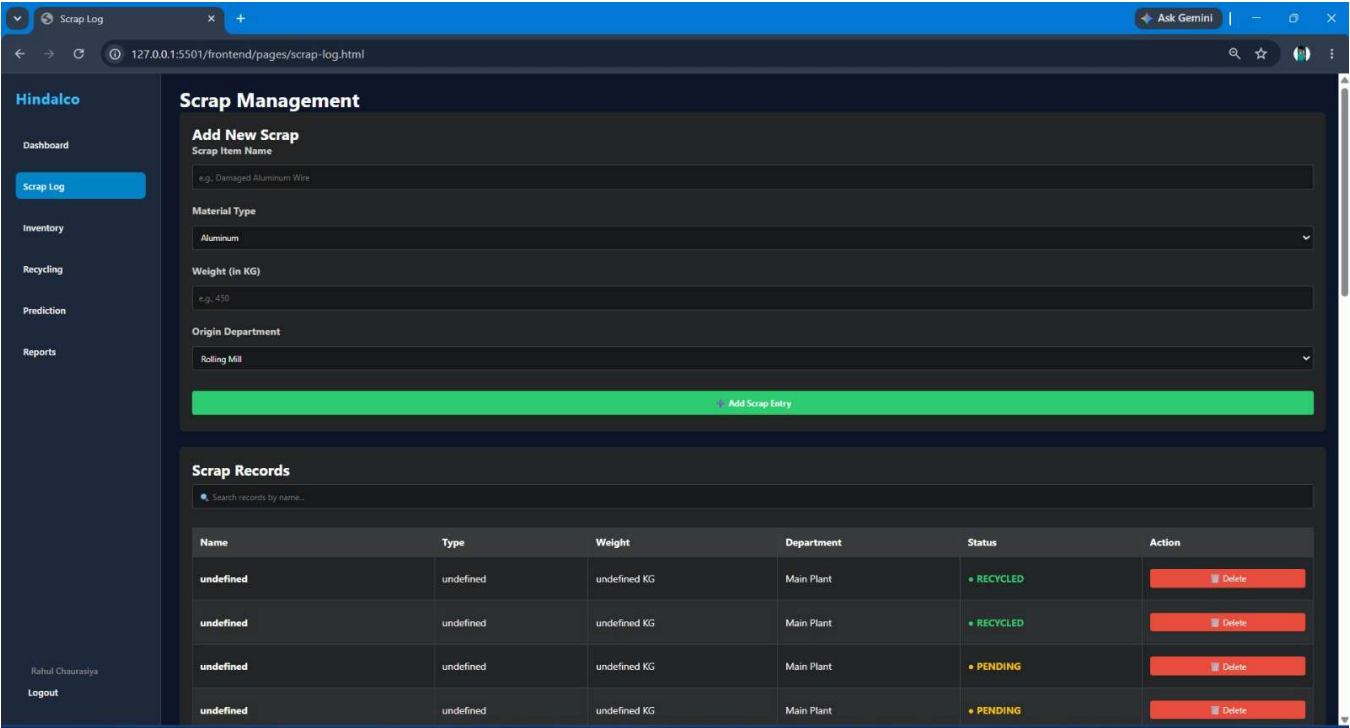


Figure: Scrap Entry Collection

This module allows users to record newly generated scrap by entering details such as scrap type, weight, department, date, and remarks. All records are stored in the database for future analysis.

This screen displays all scrap entries in a structured table format. Users can view, search, update, or manage scrap records efficiently.

2. Inventory Management

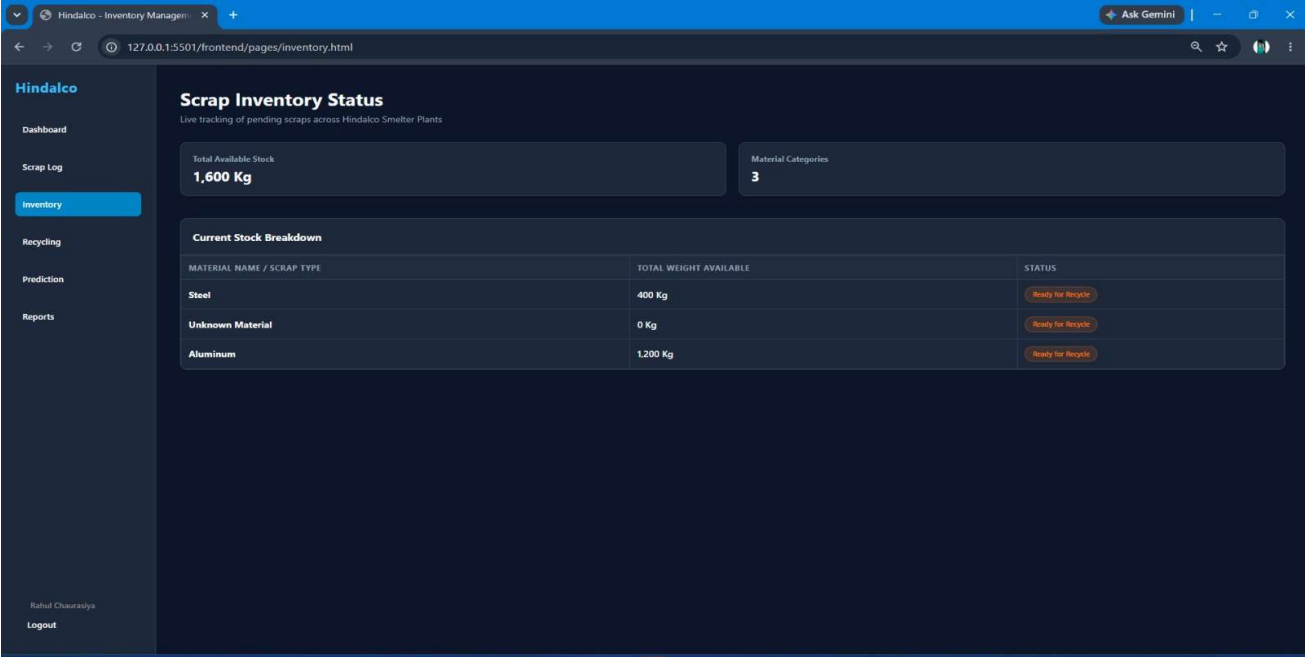


Figure: Inventory Management

The inventory module provides real-time visibility of available scrap stock. It helps track material quantities and supports effective inventory control.

6. Recycling Management

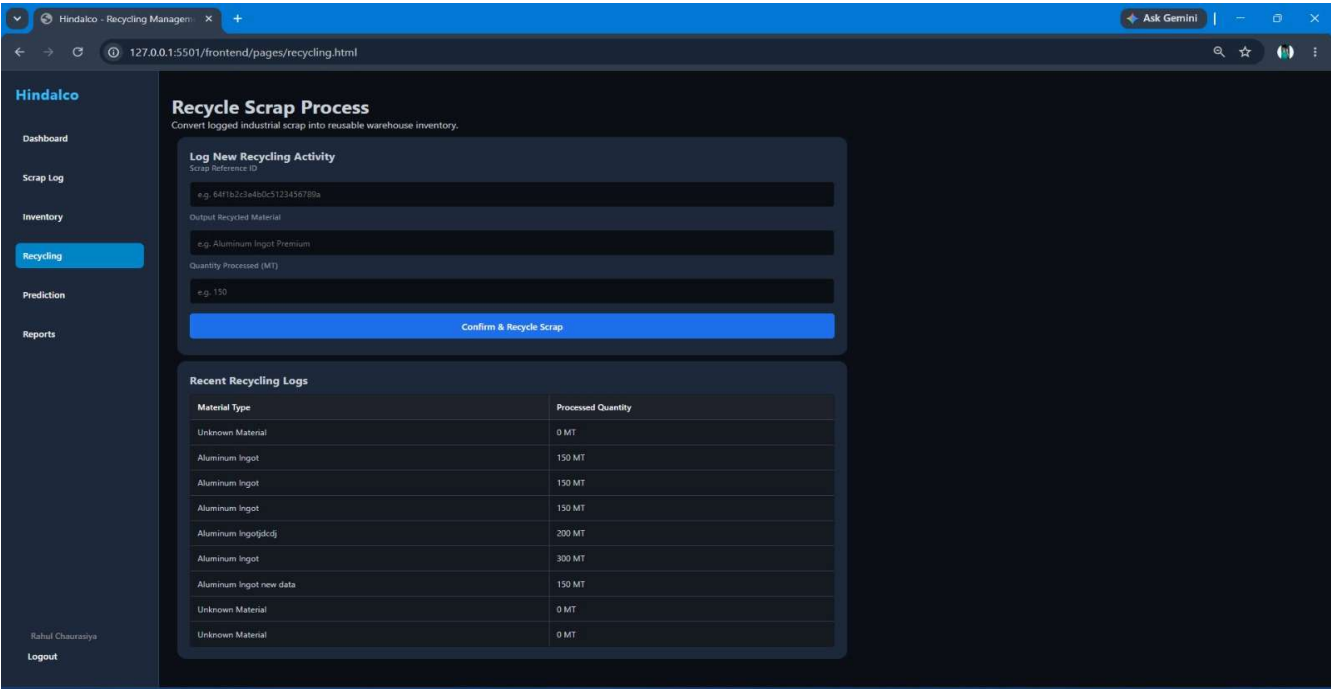


Figure: Recycling Management

This screen records recycling activities and tracks recycled as well as pending scrap quantities. It assists in evaluating recycling performance and resource utilization.

7. AI Prediction Module

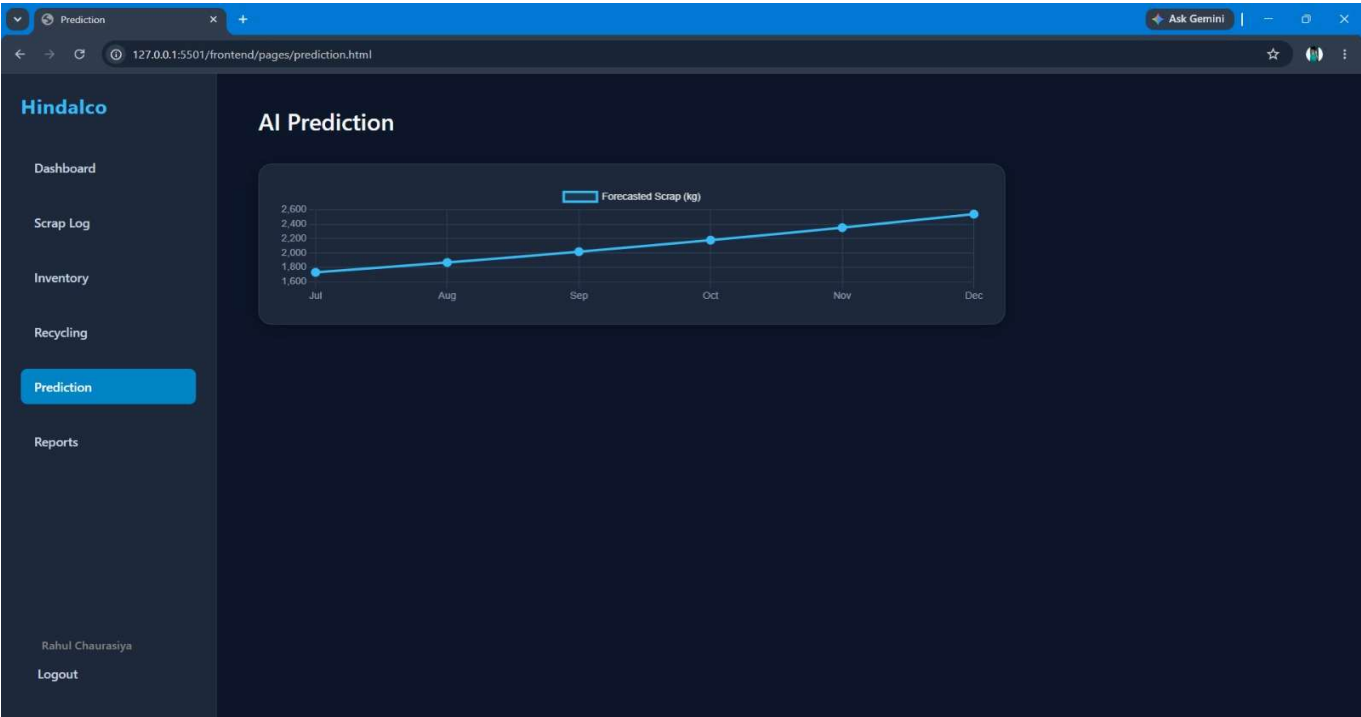


Figure: AI Prediction

The AI prediction module analyses historical scrap data and forecasts future scrap generation trends. It helps identify potential high-waste areas and supports proactive planning.

8.PDF Report Generation

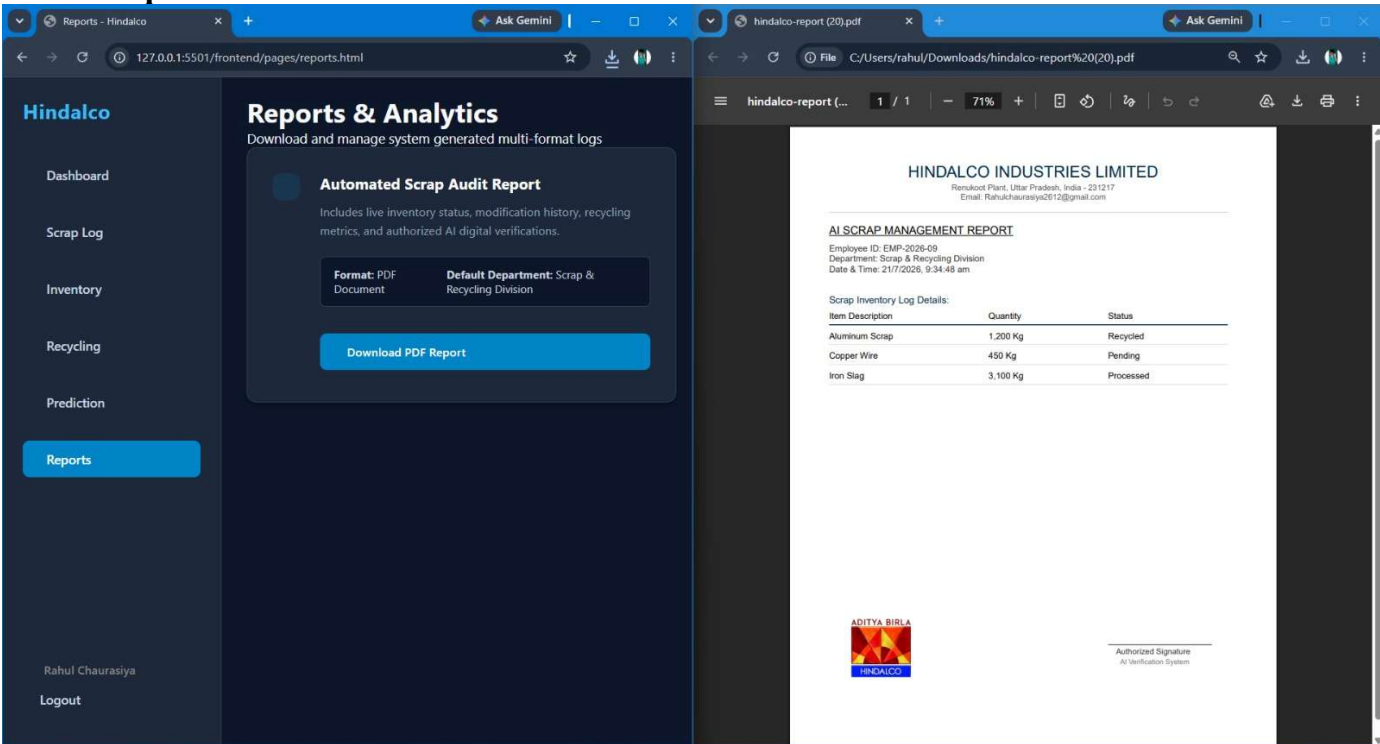
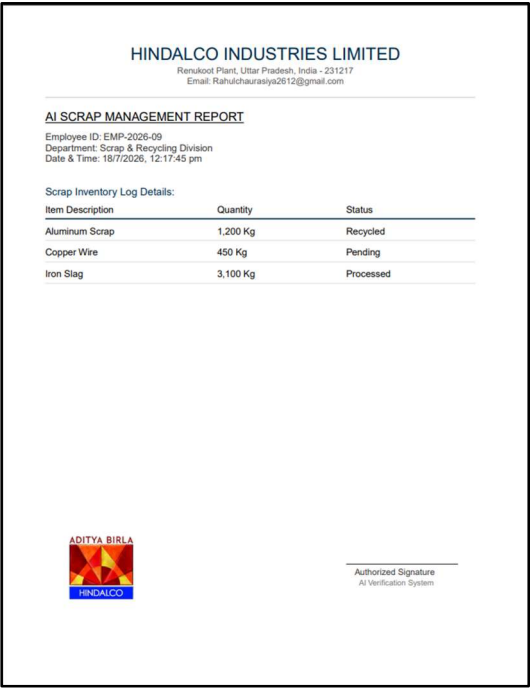


Figure: AI Prediction

This module generates downloadable PDF reports containing the latest scrap management records. The report includes scrap details, inventory status, recycling information, and summary statistics. The generated document helps management analyse operational performance and maintain proper documentation.

The PDF report consolidates all updated records available in the system database. It provides a structured summary of scrap transactions, recycling data, inventory information, and analytical insights for effective monitoring and decision-making



Software Engineering Best Practices

- Followed a **Layered Architecture** approach to ensure modular and maintainable system design.
- Implemented **Database Normalization** to reduce data redundancy and improve data consistency.
- Applied **Input Validation** techniques to ensure accurate and reliable data entry.
- Used **Exception Handling** mechanisms to improve system stability and error management.
- Maintained **Code Reusability** through modular and structured development practices.
- Followed **Standard Coding Conventions** to enhance code readability and maintainability.
- Implemented **Role-Based Access Control** to improve system security and user management.
- Performed **System Testing and Validation** to ensure functional correctness and reliability.
- Utilized **Efficient Database Queries and Indexing** for better application performance.
- Prepared the system for **Future Scalability and Enhancements**, including AI integration and advanced analytics features.

Conclusion

The **AI-Based Scrap Management and Recycling Optimization System for Hindalco Industries** provides a centralized solution for managing scrap generation, inventory tracking, recycling activities, and data analysis. The system improves operational efficiency, reduces material loss, and supports better decision-making through real-time monitoring and reporting.

By incorporating AI-based prediction and analytics, the project contributes to improved resource utilization and sustainable manufacturing practices, making it a valuable solution for industrial waste management.

Future Scope

- AI-based advanced scrap prediction
- Computer Vision for scrap classification
- QR Code-based scrap tracking
- Mobile application integration
- ERP system integration
- Automated alerts and notification